Ex:No: 1H Date: 04.02.26	Travel Itinerary Management Using Lists

Problem Description:

A travel agency wants to maintain **custom travel itineraries for multiple clients**. Each itinerary includes:

- **Client Name** (string)
- **Itinerary ID** (string)
- **Destinations** (vector of city names)
- **Days Spent** (numeric vector, corresponding to each destination)
- **Mode of Transport** (vector of strings: Flight, Train, Bus)

The agency wants to:

1. **Store each client's itinerary as a list**, allowing heterogeneous data.
2. **Calculate total travel days** for each client.
3. Identify clients whose **total travel exceeds a user-defined limit** for special assistance.

This allows the agency to manage **multiple itineraries dynamically** and perform **quick analyses** on travel plans.

Objective:

To use lists in R to store heterogeneous travel itinerary data, calculate total travel days, and identify clients with extended trips for special planning or assistance.

Sample Input	
Enter number of clients: 2 Enter maximum recommended travel days: 10 Enter details for Client 1 : Enter client name: Des1 Enter itinerary ID: A123 Enter number of destinations: 3 Enter destination 1 name: Des1 Enter number of days in Des1 : 6 Enter mode of transport to Des1 : Flight Enter destination 2 name: Des2 Enter number of days in Des2 : 3 Enter mode of transport to Des2 : Car Enter destination 3 name: Des3 Enter number of days in Des3 : 2 Enter mode of transport to Des3 : Car	Enter details for Client 2 : Enter client name: Aaron Enter itinerary ID: A321 Enter number of destinations: 2 Enter destination 1 name: Des4 Enter number of days in Des4 : 4 Enter mode of transport to Des4 : Flight Enter destination 2 name: Des5 Enter number of days in Des5 : 5 Enter mode of transport to Des5 : Car

Expected Output	
Travel Itineraries: [[1]] [[1]]$Client [1] "Des1" [[1]]$ID [1] "A123" [[1]]$Destinations [1] "Des1" "Des2" "Des3" [[1]]$Days [1] 6 3 2 [[1]]$Transport [1] "Flight" "Car" "Car"	[[2]] [[2]]$Client [1] "Aaron" [[2]]$ID [1] "A321" [[2]]$Destinations [1] "Des4" "Des5" [[2]]$Days [1] 4 5 [[2]]$Transport [1] "Flight" "Car" Clients with total travel days exceeding 10 : Client: Des1 | Itinerary ID: A123 | Total Days: 11

Program:

```
# User input num_clients and max_days
num_clients <- as.integer(readlist("Enter number of clients:"))
max_days <- as.integer(readlist("Enter maximum
                                recommended travel days: "))


# Initialize list itineraries
itineraries <- list()
```

INPUT:

Enter number of clients: 1
Enter maximum recommended travel days: 5
Enter details for client 1:
Enter client name: Anu
Enter itinerary ID: B101
Enter number of destinations: 2
Enter destination 1 name: Kochi
Enter number of days in Kochi: 3
Enter mode of transport in Kochi: Train
Enter destination 2 name: Trivandrum
Enter number of days in Trivandrum: 4
Enter mode of transport to Trivandrum: Bus

OUTPUT:

Travel Itineraries:
[[1]]
[[1]] $ client
[1] "Anu"
[[1]] $ ID
[1] "B101"
[[1]] $ Destinations
[1] "Kochi" "Trivandrum"
[[1]] $ Days
[1] 3 4
[[1]] $ Transport
[1] "Train" "Bus"
clients with total travel days exceeding 5:
client: Anu / Itinerary ID: B101 / Total days: 7

Input client itineraries

```
for (i in 1: num_clients) {
  cat ("\n Enter details for client", i, ":\n")
  client_name <- readline ("Enter client name: ")
  itinerary_id <- readline ("Enter itinerary ID: ")
  num_dist <- as.integer (Readline ("Enter number of
                                    destinations : "))
  destinations <- c()
  days <- c()
  transport <- c()
  for (j in 1: num_dest) {
    dest_name <- readline (paste ("Enter destination ", j, "name:"))
    dest_days <- as.numeric (readline (paste ("Enter number of
                                  days in ", dest_name, ": ")))
    dest_transport <- readline (paste ("Enter mode of transport
                                  to", dest_name, ": "))
    destinations <- c(destinations , dest_name)
    days <- c (days, dest_days)
    transport <- c (transport, dest_transport)
  }
  Itineraries [[i]] <- list (
      Client = client_name, ID = itinerary_id, Destinations=
        destinations, Days = days, Transport = transport
  )
}
```

```
# Display all itineraries
cat ("in Travel itineraries :\n")
print (itineraries)


# Identify clients exceeding travel limit
cat ("\n clients with total travel days exceeding",
                    max_days, ":\n")
for (i in 1 : length (itineraries)) {
  total_days <- sum (itineraries [[i]]$Days)
  if (total_days > max_days) {
    cat (
      "Client : ", itineraries [[i]]$ client,
      "\itinerary ID : ", itineraries [[i]]$ ID,
      "\ Total days: ", total_days, "\n"
    )
  }
}
}
```

Result:

Thus, the program successfully demonstrates the use of lists in R to store heterogeneous travel itinerary data, calculate total travel days per client, and identify clients requiring special assistance due to extended travel.